AeroTrace | Demonstration ICD | AT-DEMO-ICD-001 revision 1

SYNTHETIC INPUT ONLY - not an aircraft or hardware acceptance ICD.

MIL-STD-1553: RT 3, SA 2, BC-to-RT (Rx), 8 data words. Word indices are zero-based.

Unsigned binary fields; LSB 0 is the least-significant bit. Engineering = extracted integer * scale + offset.

MSW first. Rules match either channel and bus; select a single channel/bus when plotting.

Name	RT	SA	Direction	Word index (0-based)	LSB	Width	Scale	Offset	Unit	Type	Word order
TEMP	3	2	Rx	0	0	16	0.01	0	degC	uint16	MSW first
PRESSURE	3	2	Rx	1	0	16	0.01	0	kPa	uint16	MSW first
VOLTAGE	3	2	Rx	2	0	16	0.01	0	V	uint16	MSW first
VIBRATION	3	2	Rx	3	0	16	0.01	0	g	uint16	MSW first
RPM	3	2	Rx	4	0	16	0.1	0	RPM	uint16	MSW first
CURRENT	3	2	Rx	5	0	16	0.01	0	A	uint16	MSW first
PITCH	3	2	Rx	6	0	16	0.01	0	deg	uint16	MSW first
STATUS	3	2	Rx	7	0	16	1	0	bits	uint16	MSW first
READY	3	2	Rx	7	3	1	1	0	state	uint16	MSW first

Worked payload (hex): 03E8 1A2B 0B22 00F2 8313 01A2 04DD 0008

Expected: TEMP 10 degC; PRESSURE 66.99 kPa; VOLTAGE 28.5 V; VIBRATION 2.42 g;

RPM 3355.5; CURRENT 4.18 A; PITCH 12.45 deg; STATUS 8; READY 1 (word 7 bit 3).

TEMP is temperature measured in degC. PRESSURE is hydraulic pressure measured in kPa.

Time display: HH:MM:SS.mmm.uuu. It is relative device time, not automatically UTC.

Repeated timestamps can be valid at the configured card resolution; gaps must not be filled or hidden.

PDF ingestion builds a local evidence index and conversion profile. It does not retrain model weights.